ENGR 1451/2451: Exploratory Data Science

Decision Errors, Power, and Classification Metrics

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Today: What Can Go Wrong with a Decision?

Part 1: A familiar die example

- Compare **100 vs. 1,000 rolls** of a die.
- Choose a false-alarm threshold α and weigh mistakes.
- Distinguish **two decisions and four outcomes**, β, and **power**.

Part 2: Apply the same reasoning

- Connect decisions to **sensitivity**, **specificity**, and **accuracy**.
- Apply the framework to **material-strength screening**.

Recall Previous Example: Is the Die Fair?

Fair-die model

Each face $1, 2, 3, 4, 5, 6$ is equally likely.

$$\mu_0 = 3.50, \qquad \sigma_0 = \sqrt{35/12} \approx 1.708.$$

Roll the die **100 times** and add the spots.

What was observed?

$$\text{Total} = 368, \qquad \bar{x} = \frac{368}{100} = 3.68.$$

The observed mean is 0.18 above the fair-die value.

Hypotheses: H_0: the die is fair. H_A: the die is not fair.

Compare the discrepancy with the variation expected by chance.

Review: Did the Die Give Convincing Evidence?

Compare the mean with its SE

$$\mathrm{SE}(\bar{X}) = \frac{1.708}{\sqrt{100}} \approx 0.171,$$

$$z = \frac{3.68 - 3.50}{0.171} \approx 1.05.$$

The average is about **1.05 SEs** above 3.50.

Two-sided test from w09a

$$P \approx 2\,\mathrm{P}(Z \geq 1.05) \approx 0.29.$$

At significance level $\alpha = 0.05$, this is **not enough evidence** to flag the die.

Not flagging means insufficient evidence, not proof of fairness.

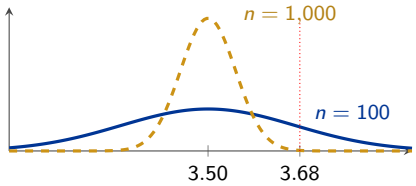
Same Average, More Evidence: 1,000 Rolls

Suppose we instead roll the die **1,000 times** and observe **3,680 spots**. The average is still $\bar{x} = 3.68$ (a difference of 0.18 from 3.50).

Smaller SE with more rolls

$$\mathrm{SE}(\bar{X}) = \frac{1.708}{\sqrt{n}}$$

	$n = 100$	$n = 1{,}000$
SE	0.171	0.054
z	1.05	3.33
Two-sided P	0.292	0.00086



A Decision Rule: Where Do the Cutoffs Come From?

Choose $\alpha = 0.05$ before rolling

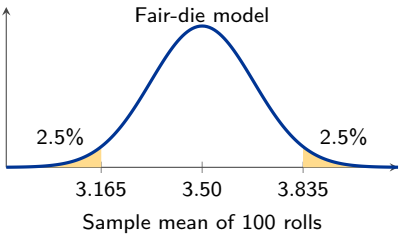
For a two-sided normal approximation:

$$\text{Flag if } |z| > 1.96.$$

For $n = 100$ independent rolls,

$$3.5 \pm 1.96\sqrt{\frac{35/12}{100}} \approx [3.165,\ 3.835].$$

Flag outside; do not flag inside.



At 100 rolls, 3.68 **is inside the cutoffs; at 1,000 rolls, it is outside.**

Each Decision Can Be Correct or Incorrect

Case 1: 100 Rolls

$P \approx 0.29 > \alpha = 0.05$

Do not flag the die

Two possible realities:

- Die is fair: **True Negative (TN)**
- Die is loaded: **False Negative (FN)**

Case 2: 1,000 Rolls

$P \approx 0.0009 < \alpha = 0.05$

Flag the die

Two possible realities:

- Die is fair: **False Positive (FP)**
- Die is loaded: **True Positive (TP)**

The sample determines the decision; the truth determines whether it is correct.

Two Decisions $\times$ Two Realities = Four Outcomes

Positive means flagged; **negative** means not flagged.

Actual condition	**Do not flag**	**Flag**
Die is fair (H_0 true)	Correct: **true negative** (TN)	Error: **false positive** (FP)
Die is loaded (H_0 false)	Error: **false negative** (FN)	Correct: **true positive** (TP)

We usually do not know whether H_0 or H_A is true, but we must consider both possibilities.

Two Kinds of Error: α and β

Type I = false positive

Flag a die that is actually **fair**.

$$\boxed{\alpha = \text{P(flag} \mid \text{fair)}}$$

For the original rule, $\alpha \approx 0.05$.

About **5 of 100** fair-die experiments would be flagged.

Type II = false negative

Fail to flag a die that is actually **loaded**.

$$\boxed{\beta = \text{P(do not flag} \mid \text{loaded)}}$$

The value of β depends on *how* the die is loaded.

α: false alarms among fair dice. **β: misses among loaded dice.**

Hypothesis Testing: Criminal Trial

Think of a criminal trial as a hypothesis test.

Hypotheses

H_0 : Defendant is innocent

H_A : Defendant is guilty

Which type of error occurs in each situation?

1. The defendant is found **not guilty**, but is actually guilty.
2. The defendant is found **guilty**, but is actually innocent.
3. Which error do you think is the worse error to make?

Is a 5% False-Alarm Rate Too High?

Suppose a manufacturer tests **1 million genuinely fair dice**, once per die.

Approximate expected false flags

Significance α	Fair dice flagged
5%	50,000
1%	10,000
0.1%	1,000
0.01%	100

Choose before collecting data

A smaller α means **fewer false alarms**.

But a stricter rule can **miss more loaded dice** (increase β).

The choice depends on the costs of mistakes and follow-up testing.

Small α reduces false alarms, but may increase misses at fixed sample size.

Discussion: Decisions and False Alarms

Use the fair-die experiment

1. A **fair die** is flagged. Which error has occurred?
2. True or false: $\alpha = 0.05$ means 5% of *flagged dice* are fair. Explain which group belongs in the denominator.

To Calculate β, Specify a Loaded Die

One possible loaded die

Imagine that rolling a **6 is more likely**:

$$\mathrm{P}(6) = \frac{1}{3}, \quad \mathrm{P}(1) = \cdots = \mathrm{P}(5) = \frac{2}{15}.$$

Then

$$\mu_1 = 4.00, \quad \sigma_1 = \sqrt{10/3} \approx 1.826.$$

Same rule, new truth

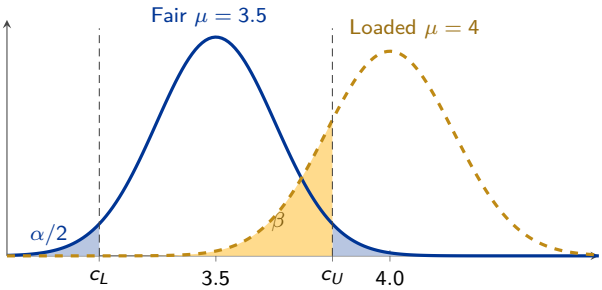
Return to $\alpha = 0.05$: roll $n = 100$ times and flag if

$$\bar{X} < 3.165 \quad \text{or} \quad \bar{X} > 3.835.$$

But now the sampling distribution is centered at **4.00**, not 3.50.

Its shifted sampling distribution makes exceeding the upper cutoff more likely.

One Decision Rule, Two Sampling Distributions



Sample mean of 100 rolls

Read each area

$\alpha \approx 0.05$
Both outer tails under the **fair** model. $\beta \approx 0.183$

Middle region under this **loaded** model.

Gold middle: loaded-die experiments that are not flagged (Type II errors).

Power: How Often Do We Catch a Loaded Die?

Power is the detection rate

For our specified loaded die,

$$\boxed{\text{Power} = 1 - \beta \approx 0.817.}$$

Across **100 repeated tests** of that die, expect roughly

- **82** flagged (correct).
- **18** not flagged (missed).

Detection terminology

$$\text{Sensitivity} = 1 - \beta \approx 82\%.$$

$$\text{Specificity} = 1 - \alpha \approx 95\%.$$

Sensitivity is for **loaded** dice
Specificity is for **fair** dice.

A test may have both false alarms (α) and missed detections (β).

Specificity, Sensitivity, and Accuracy

Specificity: among fair dice

What fraction are correctly **not flagged**?

$$\text{Specificity} = \frac{TN}{TN + FP} \approx 1 - \alpha$$

Sensitivity: among loaded dice

What fraction are correctly **flagged**?

$$\text{Sensitivity} = \frac{TP}{TP + FN} \approx 1 - \beta$$

Accuracy: among all dice

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy also depends on how common the loaded dice are.

Stricter Rules

Keep $n = 100$ and the same fair and loaded models.

False-positive rate α	False-negative rate β	Power $1 - \beta$
5%	18%	82%
1%	37%	63%
0.1%	63%	37%
0.01%	82%	18%

Tighter rule

Reducing α means fewer false positives, **but** more false negatives.

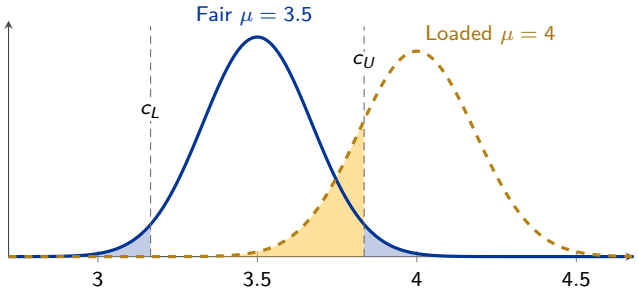
How else can we improve power?

Collect more rolls. At the same $\alpha = 0.05$:

$$n = 100: \ 82\%, \qquad n = 400: > 99\%.$$

Smaller α: Fewer False Alarms, More Misses

Same two models, $n = 100$. Flag means outside the two dashed cutoffs.



Cutoff: $\alpha = 5\%$

$\beta \approx 18\%$

Power $\approx 82\%$

Blue tails: α

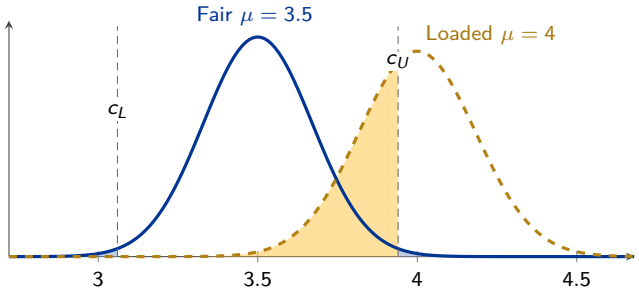
Gold middle: β

Do not flag between cutoffs; **flag** outside.

Takeaway: As α decreases, the cutoffs move outward. The blue false-alarm area shrinks, while the gold missed-detection area grows.

Smaller α: Fewer False Alarms, More Misses

Same two models, $n = 100$. Flag means outside the two dashed cutoffs.



Cutoff: $\alpha = 1\%$

$\beta \approx 37\%$

Power $\approx 63\%$

Blue tails: α

Gold middle: β

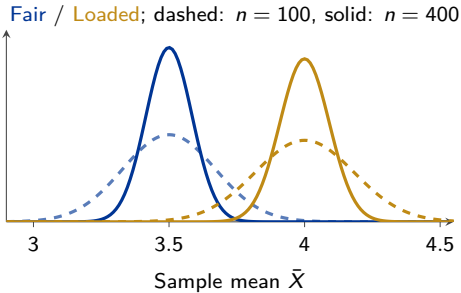
Do not flag between cutoffs; **flag** outside.

Takeaway: As α decreases, the cutoffs move outward. The blue false-alarm area shrinks, while the gold missed-detection area grows.

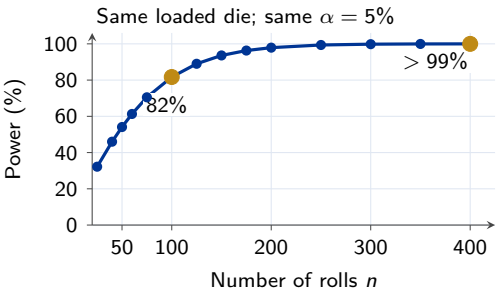
More Rolls: Narrower Curves, Higher Power

Keep $\alpha = 0.05$ **fixed.** With more independent rolls, $\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$ gets smaller.

Sampling distributions of $\bar{X}$



Power increases with n



Takeaway: Four times as many rolls **halve the SE**, so the same loaded die is much easier to detect.

Discussion: Cutoff or More Rolls?

Same specified loaded die

1. With $n = 100$, tightening α from 5% to 1% changes power from 82% to 63%. Why?
2. At the same $\alpha = 0.05$, increasing rolls from $n = 100$ to $n = 400$ changes the SE and power. In which direction?

Why Accuracy Can Be Misleading

Suppose only **0.1% of dice are loaded**.

Our screening rule

Sensitivity $\approx 82\%$; specificity $\approx 95\%$.

$$\text{Accuracy} \approx (.001)(.82) + (.999)(.95)$$

$$\boxed{\text{Accuracy} \approx 95.0\%.}$$

Always say "fair"

Never flag any die.

$$\text{Sensitivity} = 0\%,$$

$$\boxed{\text{Accuracy} = 99.9\%.}$$

It misses **every** loaded die.

High accuracy can hide a failure to detect rare loaded dice.

The Base-Rate Fallacy

Suppose we test **1 million dice**: 1,000 loaded (0.1%) and 999,000 fair.

Use the same **82% sensitivity** and **95% specificity**.

Approximate expected counts

	Loaded	**Fair**
Flagged	820	49,950
Not flagged	180	949,050

Positive predictive value: among flagged dice

What fraction are actually loaded?

$$\text{PPV} = \frac{TP}{TP + FP} = \frac{820}{820 + 49{,}950}$$

$$\boxed{\text{P(loaded} \mid \text{flagged)} \approx 1.6\%}$$

Sensitivity asks about loaded dice. Positive predictive value asks about flagged dice.

Most alerts are false alarms because fair dice are much more common.

Base-Rate Fallacy: More Examples

The base rate is how common the condition is in the population being tested.

Rare disease screening

A positive result is **not** the same as a confirmed diagnosis.

If the disease is rare, many more people without it are available to generate false alarms.

Security screening

An alarm is **not** the same as a confirmed threat.

If prohibited items are rare, false alarms can outnumber correctly detected items.

Interpreting an alert requires the base rate and the test's specificity.

Increasing Specificity: Fewer False Alarms

Recall **999,000 fair dice** and **1,000 of our specified loaded type**.

Rolls / die	α	Specificity	False flags	Power
100	5%	95%	$\approx$ 49,950	$\approx$ 82%
100	0.1%	99.9%	$\approx$ 999	$\approx$ 37%
1,000	0.1%	99.9%	$\approx$ 999	$>$ 99.9%

Smaller α reduces false alarms; more rolls can recover detection power.

Last row: specificity is not positive predictive value

$$\text{PPV} \approx \frac{1{,}000}{1{,}000 + 999} \approx 50.0\%.$$

Two-Stage Screening: What Can It Improve?

Idea: Screen every die, but perform additional testing only on suspicious dice.

Stage 1: Initial Screening

- Test every die (e.g., 100 rolls).
- Flag potentially loaded dice.
- Some loaded dice will be missed.

Stage 2: Confirmation

- Retest only flagged dice.
- Use more rolls or a different method.
- Reject only if confirmed.

Confirmation can improve specificity, but cannot increase overall sensitivity.

If Stage 1 catches 82% of loaded dice, the final sensitivity is at most 82%.

Discussion: Rare Events and Follow-up Tests

Consider the loaded-die screening system

1. With only 0.1% loaded dice, sensitivity is 82%, but positive predictive value is just 1.6%. Why?
2. If Stage 1 detects 82% of loaded dice and only flagged dice reach Stage 2, can final sensitivity reach 99%?

Different Loading Patterns Have Different Power

Our die is a challenge—about **18% missed**—but **not the hardest possible case**.

Keep $n = 100$, $\alpha = 0.05$, and the same two-sided mean test.

Illustrative loading	P(6)	True mean	Power (approx.)
Smaller mean shift	1/4	3.75	32%
Our example	1/3	4.00	82%
Larger mean shift	1/2	4.50	$> 99.9\%$

In each row, faces 1–5 share the remaining probability equally.

Larger shifts are easier to detect here; smaller shifts are harder.

Important limitation: Some unfair dice still have mean 3.5. Check all face frequencies to assess general fairness.

Testing a Manufacturer's Claim

Same decision, a new setting

A manufacturer claims a batch has **mean tensile strength of at least 50 MPa**.

$$H_0 : \mu \geq 50, \qquad H_A : \mu < 50.$$

Flag the batch if there is convincing evidence that its mean is too low.

Hypothetical sample measurements

Randomly select **25 specimens** from the batch.

$$n = 25, \qquad \bar{x} = 48 \text{ MPa},$$

$$\text{sample SD:} \quad s = 5 \text{ MPa}.$$

The population mean μ and SD σ are **unknown**.

Unlike the specified die model, we must estimate variability from the sample.

Assumptions: Independent observations from an approximately normal batch population.

Material Strength: Estimate the SE, Then Use t

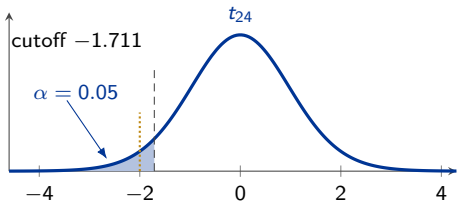
Use the sample SD

$$\widehat{\mathrm{SE}}(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{5}{\sqrt{25}} = 1 \text{ MPa}.$$

$$t = \frac{\bar{x} - 50}{s/\sqrt{n}} = \frac{48 - 50}{1} = -2.00.$$

$$\mathrm{df} = n - 1 = 24.$$

Reference distribution at $\mu = 50$



Gold dotted line: observed $t = -2.00$.

Choose $\alpha = 0.05$: flag if $t < -1.711$.

$$P = \mathrm{P}(T_{24} \le -2.00) \approx 0.028 < 0.05 \quad \Longrightarrow \quad \boxed{\text{Flag the batch.}}$$

Why the Observed P-Value Is Not β

For the sample: $\bar{x} = 48$, $s = 5$, and $P \approx 0.028$.

What the sample provides

An estimate of the mean and variability, and evidence against the claimed strength.

The observed P-value uses the **null** reference distribution.

What a power calculation needs

A **specified true mean below 50 MPa**, a variability model, and the planned testing rule.

β describes missed detections under that **alternative**.

A numerical β cannot be determined from this sample alone.

A P-value is not a false-negative probability.

Which Error Matters More? Material Strength

False positive: good batch flagged

- Additional testing or inspection.
- Production delays.
- Possibly scrapping material unnecessarily.

False negative: weak batch not flagged

- Weak material reaches a customer.
- Reliability or safety failures.
- Warranty, replacement, or recall costs.

Choose a cutoff using error costs, prevalence, and options for retesting.

Which Error Matters More? Other Applications

False positive: false alarm

- **Rare disease:** Person without the disease tests positive; anxiety and unnecessary tests.
- **Criminal justice:** Innocent person convicted; wrongful punishment.
- **Basement water sensor:** Alarm sounds without a leak; unnecessary inspection.

False negative: missed problem

- **Rare disease:** Person with the disease tests negative; delayed treatment.
- **Criminal justice:** Guilty person not convicted; possible further offenses.
- **Basement water sensor:** Leak goes undetected; flooding and property damage.

The more serious error depends on its consequences and available follow-up.

Discussion: Apply the Material Test

Material sample: $n = 25$, $\bar{x} = 48$, $s = 5$ MPa, $P \approx 0.028$

1. Would you flag the batch at $\alpha = 0.05$? At $\alpha = 0.01$?
2. Is $\beta = 0.028$? What extra information is needed to estimate it?
3. For a basement leak sensor, which error is especially concerning, and why?

Summary: From a Die to Material Strength

Two errors

$$\alpha = \mathrm{P}(\text{flag} \mid H_0 \text{ true})$$

$$\beta = \mathrm{P}(\text{do not flag} \mid H_1 \text{ case})$$

- α: false-positive rate.
- β: false-negative rate for a specified alternative.

Performance metrics

$$\text{Specificity} = 1 - \alpha$$

$$\text{Sensitivity} = \text{Power} = 1 - \beta$$

$$\text{Accuracy} = \frac{TP + TN}{N}$$

Positive predictive value (PPV): among flags, what fraction are real? $\dfrac{TP}{TP + FP}$

Choose the rule before testing; consider both errors and the base rate.